

If the lightest neutrino mass is the dark-energy scale: a narrow window for neutrinoless double-beta decay

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(Dated: August 2, 2026)

The dark-energy density fixes a mass scale $\rho_\Lambda^{1/4} \simeq 2.25 \text{ meV}$, of the same order as the neutrino mass splittings. Numerical proximity is not evidence; we only map consequences. Taking the sharpest arithmetic reading as a scenario — $m_1 = 2.25 \text{ meV}$ with normal ordering — and evaluating the standard $m_{\beta\beta}$ map at that single point gives three contributions (1.52, 2.67, 1.10) meV, $m_{\beta\beta} \in [0.04, 5.30] \text{ meV}$, and $\Sigma m_\nu = 61.3 \text{ meV}$. The lower edge is an unprotected 1.7% accident: it exists only because the middle term exceeds the other two by 0.045 meV and vanishes entirely for $m_1 \gtrsim 2.32 \text{ meV}$; it has no observational weight. The nearer referee is cosmology: the mass sum sits just inside current combined limits and will be graded there first. Laboratory discrimination is secondary. Minimal normal ordering cannot exceed 3.69 meV, while this scenario reaches 5.30 meV and occupies the gap between them 32% of the time. One planned experiment overlaps that gap only at the favourable end of its matrix-element range, and only 10.8% of phases put $m_{\beta\beta}$ above the corresponding threshold; a detection there would favour the scenario over minimal ordering, while a detection above 5.30 meV excludes it. A null result constrains nothing, at any sensitivity. The $m_{\beta\beta}$ band itself is textbook; only the placement at this m_1 is at issue.

I. INTRODUCTION

The energy density of dark energy, expressed as a mass scale, is

$$\rho_\Lambda^{1/4} \simeq 2.25 \text{ meV}, \quad (1)$$

which lies within an order of magnitude of $\sqrt{\Delta m_{21}^2} \simeq 8.6 \text{ meV}$ and two orders of $\sqrt{\Delta m_{31}^2} \simeq 50 \text{ meV}$. That numerical proximity has been remarked on for decades and has motivated model building — most directly mass-varying-neutrino scenarios that tie the dark-energy density to the neutrino sector [1, 2]. Proximity of scales is not evidence of a relation. We do not treat it as one, propose no mechanism, and take no position on those that exist.

What follows is narrower: a single arithmetic scenario and its map. Equation (1) has one sharp reading,

$$m_1 = \rho_\Lambda^{1/4} = 2.25 \text{ meV}, \quad \text{normal ordering}, \quad (2)$$

with m_1 the lightest neutrino mass. We ask only what Eq. (2) implies once the standard oscillation inputs are fixed. No cosmological model enters; nothing computed below depends on why the equality might hold. Numerical proximity is not evidence; we only map consequences.

The map from the lightest mass to the effective Majorana mass $m_{\beta\beta}$ is textbook, and the familiar lobster-shaped allowed region in the $(m_{\text{lightest}}, m_{\beta\beta})$ plane appears in every review of neutrinoless double-beta decay [3, 4]. Evaluating that map at one particular m_{lightest} is just a new calculation. The only claim is the placement: just above the hard ceiling of minimal normal ordering,

with a mass sum that cosmology will grade before any laboratory experiment can. That placement, and the (mostly modest) detection probabilities that follow from it, are what we work out.

Section II fixes the inputs and computes the window, leading with the mass sum. Section III explains why the lower edge is nonzero and why it is an unprotected accident. Section IV sets out what would distinguish the scenario from minimal ordering and what would refute it. Section V states the assumptions.

II. THE EFFECTIVE MASS

For three Majorana neutrinos the amplitude for neutrinoless double-beta decay is controlled by

$$m_{\beta\beta} = \left| \sum_{i=1}^3 U_{ei}^2 m_i \right| \quad (3)$$

$$= \left| c_{12}^2 c_{13}^2 m_1 + s_{12}^2 c_{13}^2 m_2 e^{i\alpha_2} + s_{13}^2 m_3 e^{i\alpha_3} \right|,$$

with α_2, α_3 the Majorana phases, which oscillation experiments do not constrain. We use $\sin^2 \theta_{12} = 0.307$, $\sin^2 \theta_{13} = 0.022$, $\Delta m_{21}^2 = 7.42 \times 10^{-5} \text{ eV}^2$ and $\Delta m_{31}^2 = 2.51 \times 10^{-3} \text{ eV}^2$, consistent with current global fits [5, 6]. With $m_2 = (m_1^2 + \Delta m_{21}^2)^{1/2}$ and $m_3 = (m_1^2 + \Delta m_{31}^2)^{1/2}$, Eq. (2) gives $m_2 = 8.90 \text{ meV}$, $m_3 = 50.15 \text{ meV}$, and

$$\Sigma m_\nu = 61.3 \text{ meV}. \quad (4)$$

The three contributions to Eq. (3) are

$$(|U_{e1}^2 m_1|, |U_{e2}^2 m_2|, |U_{e3}^2 m_3|) = (1.52, 2.67, 1.10) \text{ meV}. \quad (5)$$

Varying the phases over their full range gives

$$0.04 \text{ meV} \leq m_{\beta\beta} \leq 5.30 \text{ meV} \quad (6)$$

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with $\simeq 3.3$ meV typical. The scenario fixes m_1 but says nothing about the Majorana phases, so position within Eq. (6) is unconstrained; that is a genuine limitation and not a marginal one, as Sec. IV makes quantitative.

Cosmology is the nearer referee. The mass sum in Eq. (4) is more immediately testable than anything in the laboratory programme below. Current bounds from the microwave background combined with baryon acoustic oscillations have reached $\Sigma m_\nu \lesssim 72$ meV, with some combinations pressing lower [7]. The value in Eq. (4) sits just inside those limits; if the bound continues to tighten, cosmology will grade this scenario before any $0\nu\beta\beta$ experiment can. Laboratory discrimination is secondary and is discussed only after that fact.

III. WHY THE LOWER EDGE IS NONZERO, AND WHY IT IS FRAGILE

The lower edge of Eq. (6) is not a robust floor; it is an unprotected numerical accident, and that status should be stated before any arithmetic. Complete cancellation in Eq. (3) requires the three complex terms to close a triangle, which is possible only if no one of them exceeds the sum of the other two. From Eq. (5),

$$2.67 > 1.52 + 1.10 = 2.62 \text{ meV}, \quad (7)$$

by 0.045 meV. The middle term wins, the triangle cannot close, and the lower edge follows as the residue 2.67–2.62.

That margin is only 0.045 meV on terms of order 2.6 meV — a 1.7% effect. Nothing protects it: it is not the consequence of a symmetry, and it would not survive a modest shift in θ_{12} , in Δm_{21}^2 , or in m_1 itself. At $m_1 \gtrsim 2.32$ meV the inequality reverses and the floor vanishes entirely. A 3% change in the lightest mass, or a comparably small shift in the oscillation inputs, erases the nonzero lower edge. Calling it a floor overstates what it is: an unprotected numerical accident at this particular m_1 .

The same near-degeneracy makes $m_{\beta\beta}$ a steep function of m_1 near this point. A quantity sitting close to a cancellation responds sharply to its inputs, so the band edge here is unusually sensitive to the inputs that produce it.

No observational consequence rides on the lower edge. At 0.04 meV it is two orders of magnitude below any experimental reach under discussion. Every statement in Sec. IV concerns the upper edge, which the near-cancellation barely moves.

IV. WHAT WOULD TEST IT

A. The interval that separates scenarios

Laboratory tests are secondary to the cosmological mass sum of Sec. II. Conditional on that, the comparison that matters in $0\nu\beta\beta$ is not against zero but against

the minimal alternative. Setting $m_1 = 0$ with the same ordering and parameters gives

$$m_{\beta\beta}^{\min} \in [1.48, 3.69] \text{ meV}, \quad (8)$$

so minimal normal ordering has a hard ceiling at 3.69 meV that no choice of phases can exceed. The scenario of Eq. (2) reaches 5.30 meV. The interval

$$3.69 \text{ meV} < m_{\beta\beta} < 5.30 \text{ meV} \quad (9)$$

is therefore inaccessible to minimal ordering and occupied by this scenario. Integrating Eq. (3) over flat Majorana phases, the scenario lands inside Eq. (9) 31.8% of the time.

B. Where the experiments fall

Projected ten-year reaches for the ton-scale programme, quoted as ranges because the nuclear matrix element is the dominant uncertainty, are approximately 4.7–20.3 meV for nEXO [8], 9–21 meV for LEGEND-1000 [9], and 12–34 meV for CUPID [10].

Against the ceiling of Eq. (6), LEGEND-1000 and CUPID lie entirely above the scenario and cannot reach it for any matrix element. Only nEXO overlaps, and only at the favourable end of its matrix-element range, 4.7–5.3 meV. Over flat phases the scenario exceeds 4.7 meV only 10.8% of the time — roughly one in nine. Even if Eq. (2) is correct, baseline nEXO is unlikely to see a signal: low probability is the first statement about this overlap, not a footnote to it.

Discrimination is secondary to that probability. The entire 4.7–5.3 meV overlap does lie inside Eq. (9), which minimal ordering cannot reach, so a detection there would favour the scenario over minimal normal ordering. It would not be uniquely attributable to Eq. (2): nuclear-matrix-element uncertainty, phase measure, and the secondary status of the laboratory test relative to cosmology all limit how much weight such a signal could carry.

C. Why a sensitivity upgrade does not help as much as expected

A factor-of-four improvement in half-life sensitivity, of the kind a barium-tagging upgrade is projected to deliver, corresponds to a factor $\sqrt{4} = 2$ in $m_{\beta\beta}$, taking the reach to $\simeq 2.35$ meV. The detection probability rises from 10.8% to 69.1%.

This buys likelihood, not discrimination. Minimal normal ordering also exceeds 2.35 meV, and does so 63.7% of the time. A tagged detection near that threshold would be consistent with both scenarios and would separate almost nothing. The unimproved machine is the more discriminating one in $0\nu\beta\beta$, precisely because its reach sits in the interval where the two scenarios do not overlap —

still subject to the low 10.8% probability above, and still secondary to the cosmological mass sum.

D. Falsification

A measured $m_{\beta\beta} > 5.30$ meV refutes Eq. (2) outright, since no choice of phases can produce it. Evidence for inverted ordering does the same: at $m_3 = 2.25$ meV inverted ordering gives $m_{\beta\beta} \in [18.6, 49.1]$ meV, which does not overlap Eq. (6) at any phase. Evidence that neutrinos are Dirac particles removes the observable entirely. Cosmological exclusion of $\Sigma m_\nu = 61.3$ meV would do the same, and is the nearer path: the laboratory ceiling is a secondary falsifier.

A null result in $0\nu\beta\beta$ refutes nothing. Because the phases are free, $m_{\beta\beta}$ can be arbitrarily small, and no non-observation at any laboratory sensitivity constrains Eq. (2). That asymmetry is built into the scenario.

For orientation, the strongest current limit comes from ^{136}Xe : KamLAND-Zen reports $T_{1/2} > 4.3 \times 10^{26}$ yr at 90% confidence over a 2097 kg yr exposure, corresponding to $m_{\beta\beta} < 28\text{--}122$ meV across the matrix-element range [11]. That is a factor of five or more above the ceiling of Eq. (6), so no existing laboratory measurement bears on this scenario in either direction.

V. WHAT IS ASSUMED

Majorana neutrinos. Eq. (3) presumes lepton-number violation. If neutrinos are Dirac there is no $m_{\beta\beta}$ and the question does not arise.

Normal ordering. Assumed throughout, and stated in Eq. (2) rather than derived. Global fits currently prefer it at modest significance; the preference is not settled.

Three light species. A light sterile state would add a term to Eq. (3) and widen the window.

Standard mechanism. We assume light-neutrino exchange dominates the amplitude. If other lepton-

number-violating physics contributes, $m_{\beta\beta}$ as extracted from a half-life is no longer the quantity computed here.

The numerical input. We use the observed $\rho_\Lambda^{1/4} = 2.25$ meV. Nothing depends on this at the quoted precision: shifting it by 0.5% moves the lower edge of Eq. (6) from 0.045 to 0.039 meV and leaves the upper edge at 5.30 meV, so every conclusion in Sec. IV is unchanged.

The scenario itself. Eq. (2) is an arithmetic scenario, not a result. Equating two scales because they are numerically close is not evidence; we do not argue otherwise. We write it down because it is sharp, costs nothing to state, and happens to be nearly testable — first by cosmology through Σm_ν , secondarily in the laboratory. Numerical proximity is not evidence; we only map consequences.

VI. CONCLUSION

Taking $m_1 = 2.25$ meV with normal ordering as a sharp arithmetic scenario — not as evidence drawn from the coincidence of scales — gives $m_{\beta\beta} \in [0.04, 5.30]$ meV and $\Sigma m_\nu = 61.3$ meV. The lower edge is an unprotected 1.7% accident (0.045 meV margin; vanishes for $m_1 \gtrsim 2.32$ meV) and carries no observational weight. The upper edge does the laboratory work; the mass sum is the nearer number.

Cosmology is the first referee. A sum of 61.3 meV sits just inside current combined limits, and the bound is tightening. Laboratory discrimination is secondary. Conditional on that, the useful $0\nu\beta\beta$ comparison is against minimal normal ordering, whose hard ceiling is 3.69 meV. The gap between the two ceilings is closed to minimal ordering and occupied by this scenario about a third of the time. One planned experiment overlaps that gap only at favourable matrix elements, and only 10.8% of phases put a signal above the corresponding threshold: low probability first, modest discrimination second. A detection there would favour the scenario over minimal ordering without uniquely establishing it; a detection above 5.30 meV would refute the scenario; a null laboratory result would say nothing at all.

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